7th Grade Math · Weekly Check · Unit 7 **C-AR.3.1**

Benchmark: **MA.7.AR.3.1** · Multi-step real-world percent problems Student code: _____ Calculator: yes

Learning goal: I can solve percent problems: discount, markup, tax, tip, interest, percent change, percent error. (circle your level after scoring)

4	I can solve a multi-step percent problem, including percent change or percent error, and explain the result.
3	I can solve a one-step percent problem such as a discount, tax, tip, or simple interest.
2	I can find a percent of a number.
1	I need help to start this benchmark.

1. A video game costs $76.98 and is on sale for 12% off.

What is the sale price, to the nearest cent?

2. Devon deposits $3150.00 in an account that pays 3.5% simple interest each year. Devon leaves the money in the account for 3 years, then withdraws everything.

How much does Devon withdraw?

Turn over →

Benchmark: **MA.7.AR.3.1** · Multi-step real-world percent problems Student code: _____

3. The ratio of dogs to cats among the pets at a shelter is $11:9$. Every pet at the shelter is a dog or a cat.

What percent of the pets at the shelter are dogs? If there are 180 pets at the shelter, how many are dogs?

4. The number of visitors to a museum each month changed from 320 to 344.

What is the percent increase?

5. A store takes 28% off the original price of $162.50, then takes another 32% off the reduced price.

What is the final price?

A. $117.00

B. $110.50

C. $65.00

D. $79.56

Score: ____ / 5 **My level: 1 2 3 4** **→ copy both to your progress strip**

Answer key — C-AR.3.1 · MA.7.AR.3.1 · Multi-step real-world percent problems

Unit 7: Percentage Problems With Proportional Relationships. Regular = on-benchmark item; Rigor = the item that needs the benchmark's reasoning. Mastery on the weekly check = 4 of 5.

Weekly check (5 items)

1. Regular **$67.74**

Discount: $\$76.98 \times 0.12 = \$9.2376 \approx \$9.24$.

Sale price: $\$76.98 - \$9.24 = \$67.74$.

12% must be written as 0.12 before multiplying.

2. Regular **$3480.75**

Interest: $I = Prt = 3150 \times 0.035 \times 3 = \$330.75$.

Withdrawal is the principal plus the interest: $\$3150.00 + \$330.75 = \$3480.75$.

3. Regular **55%; 99 dogs. Score: 1 point for the percent and the count; one alone earns 0.**

For every 11 dogs there are 9 cats, so 11 of every 20 are dogs.

$11 \div 20 = 0.55$, which is 55%.

55% of $180 = 0.55 \times 180 = 99$ dogs.

4. Regular **7.5%**

Change: $344 - 320 = 24$, an increase of 24.

Compare the change with the ORIGINAL amount: $24 \div 320 = 0.075$.

$0.075 \times 100 = 7.5\%$.

5. Rigor **D. $79.56**

First discount: $\$162.50 \times 72\% = \$162.50 \times 0.72 = \$117.00$.

The second discount applies to $117.00, not to the original price: $\$117.00 \times 0.68 = \$79.56$.

Taking 28% and then 32% is not the same as taking $60\%(\$65.00)$, and the second discount is not taken from the original price $(\$110.50)$.

Re-check strip (3 items, C-AR.3.1-R)

1. Regular **32.5%**

Change: $162 - 240 = -78$, a decrease of 78.

Compare the change with the ORIGINAL amount: $78 \div 240 = 0.325$.

$0.325 \times 100 = 32.5\%$.

2. Regular **$92.51**

Tax: $\$86.46 \times 0.07 = \$6.0522 \approx \$6.05$.

Total: $\$86.46 + \$6.05 = \$92.51$.

3. Regular **6.25%**

Percent error compares the size of the error with the ACTUAL amount.

Error: $|285 - 304| = 19$.

$19 \div 304 = 0.0625$.

$0.0625 \times 100 = 6.25\%$.

Comparison strip (2 items, C-AR.3.1-X) — used on a later week's re-check when this benchmark was *not* retaught

1. Regular **$2706.00**

Interest: $I = Prt = 2400 \times 0.0425 \times 3 = \$306.00$.

Withdrawal is the principal plus the interest: $\$2400.00 + \$306.00 = \$2706.00$.

2. Regular **$58.89**

Tax: $\$54.78 \times 0.075 = \$4.1085 \approx \$4.11$.

Total: $\$54.78 + \$4.11 = \$58.89$.

Reteach plan — C-AR.3.1

MA.7.AR.3.1 · Multi-step real-world percent problems

Item that gave the most trouble (from both tally sheets):

What the mistake looks like:

Worked example for Tuesday (8–10 min):

Three practice items (one change at a time):

Decision line: Reteach on _____ Huddle: in person / paper Date: _____ Re-check given on _____

7th Grade Math · Re-check (3 items)

C-AR.3.1-R

Benchmark: **MA.7.AR.3.1** · Multi-step real-world percent problems

Student code: _____ Calculator: yes

1. The number of visitors to a museum each month changed from 240 to 162.

What is the percent decrease?

2. A bike helmet costs $86.46. The sales tax rate is 7%.

What is the total cost, including tax, to the nearest cent?

3. A student estimated the number of steps on a trail to be 285. The actual number was 304 steps.

What is the percent error of the estimate?

Score: ____ / 3 → **copy to your progress strip**

7th Grade Math · Re-check (3 items)

C-AR.3.1-R

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Student code: _____ Calculator: yes

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7th Grade Math · Comparison items (2) C-AR.3.1-X

Benchmark: **MA.7.AR.3.1** · Multi-step real-world percent problems Student code: _____ Calculator: yes

1. Andre deposits $2400.00 in an account that pays 4.25% simple interest each year. Andre leaves the money in the account for 3 years, then withdraws everything.

How much does Andre withdraw?

2. A tent costs $54.78. The sales tax rate is 7.5%.

What is the total cost, including tax, to the nearest cent?

Score: ____ **/ 2** → **copy to your progress strip**

7th Grade Math · Comparison items (2) C-AR.3.1-X

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Score: ____ **/ 2** → **copy to your progress strip**